Random Variable

- ❖ A random variable is a mathematical concept used in probability theory and statistics to describe numerical outcomes that result from a random experiment, process, or phenomenon. In simpler terms, it is a variable whose values are determined by chance. Random variables are a key component in the study of probability and are used to model uncertain situations.
- ❖ There are two main types of random variables:

Discrete Random Variable:

- ❖ A discrete random variable is one that takes on a countable number of distinct values. These values are often isolated points on the number line.
- ❖ Examples of discrete random variables include the number of heads obtained when flipping a coin, the number of cars passing through a toll booth in an hour, or the number of emails received in a day.
- ❖ The probability distribution of a discrete random variable is often described using a probability mass function (PMF), which gives the probability of each possible value of the random variable.

Continuous Random Variable:

- A continuous random variable is one that can take on any value within a specified range or interval.
- Continuous random variables are associated with continuous phenomena and have an infinite number of possible values.
- Examples include the height of a person, the temperature in a room, or the time it takes for a computer to process a task.
- The probability distribution of a continuous random variable is described using a probability density function (PDF).
- Unlike the PMF for discrete random variables, the PDF does not directly give probabilities for specific values but instead provides the probability density over intervals.

The Distribution Function

- In the theoretical discussion on Random Variables and Probability, we note that the probability distribution encouraged by a random variable X is determined uniquely by a consistent assignment of mass to semi-infinite intervals of the form $(-\infty,t]$ for each real t.
- This suggests that a natural description is provided by the following.
- **Definition**
- The distribution function F_X for random variable X is given by

$$F_X(t)P(X \leq t) = P(X \in (-\infty, t]) \ \forall t \in R$$

- In terms of the mass distribution on the line, this is the probability mass at or to the left of the point t. As a consequence

Density Function

- A density function, also known as a probability density function (PDF), is a function that describes the likelihood of a random variable appearing as a certain value.
- The function's value at any given sample in the sample space can be interpreted as the relative likelihood that the value of the random variable would be equal to that sample